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Ubuntu Server Installation Guide

ABC Startup Solutions – Linux Server Deployment Project

Week 03 – BSIT Self-Paced Learning Module: System Administration and Maintenance

Prepared by: Syron B. Blaza

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Project Overview

ABC Startup Solutions is deploying its first Linux server to support file sharing, remote administration, database hosting, web hosting, and internal services. As the newly hired Junior System Administrator, this project documents the full process of installing and configuring Ubuntu Server following recommended enterprise installation practices, verifying the resulting server, and comparing it against alternative server operating systems (Windows Server and Rocky Linux) so future administrators can reproduce or extend the deployment.

Objectives

- Deploy a clean Ubuntu Server virtual machine using recommended enterprise installation practices.
- Verify server functionality after installation (login, hostname, IP, connectivity, updates, SSH).
- Compare BIOS and UEFI firmware and explain why UEFI has become the modern standard.
- Document the Linux boot process from power-on to login prompt.
- Install Windows Server Evaluation as a comparison platform.
- Compare Windows Server, Ubuntu Server, and Rocky Linux for enterprise use.
- Produce professional documentation reproducible by another administrator.

Software Requirements

- Oracle VirtualBox (virtualization platform)
- Ubuntu Server 24.04.4 LTS ISO (latest stable version at time of installation)
- Windows Server 2025 Evaluation ISO (bring-home activity)

Virtual Machine Specification

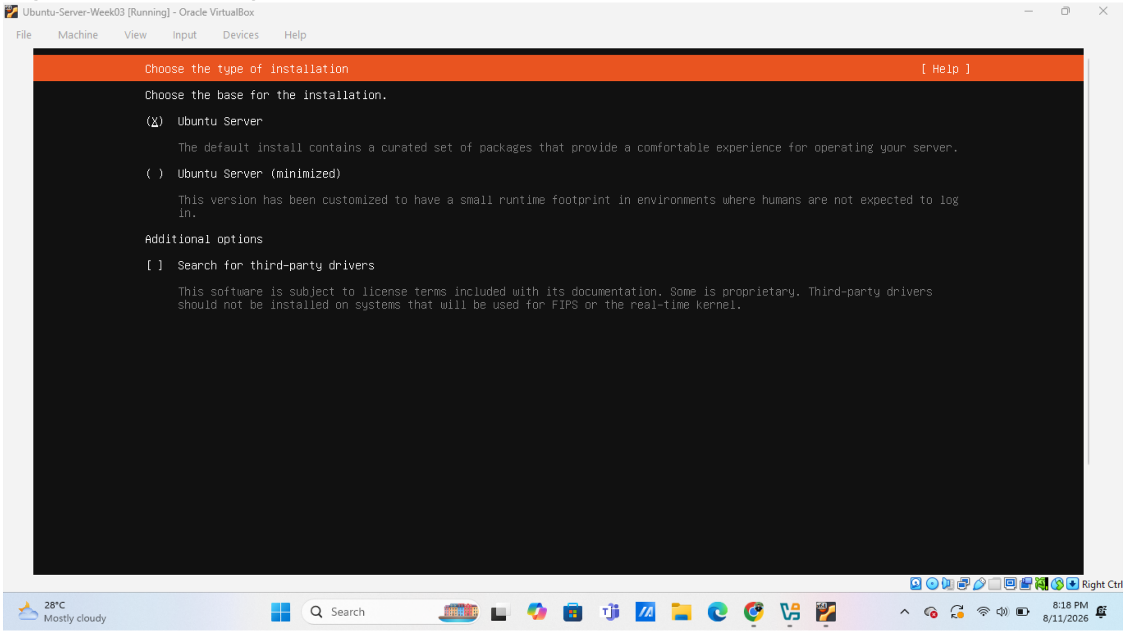
Component	
Name	Ubuntu-Server-Week03
RAM	1536 MB – 2048 MB (adjusted down from the 4 GB spec due to host hardware limits; see Problems Encountered)
CPU	2 Virtual Processors
Storage	1536 MB – 2048 MB (adjusted down from the 4 GB spec due to host hardware limits; see Problems Encountered)
Network	NAT (Intel PRO/1000 MT Desktop Adapter)

Optical Drive	ubuntu-24.04.4-live-server-amd64.iso

Step-by-Step Installation Procedure

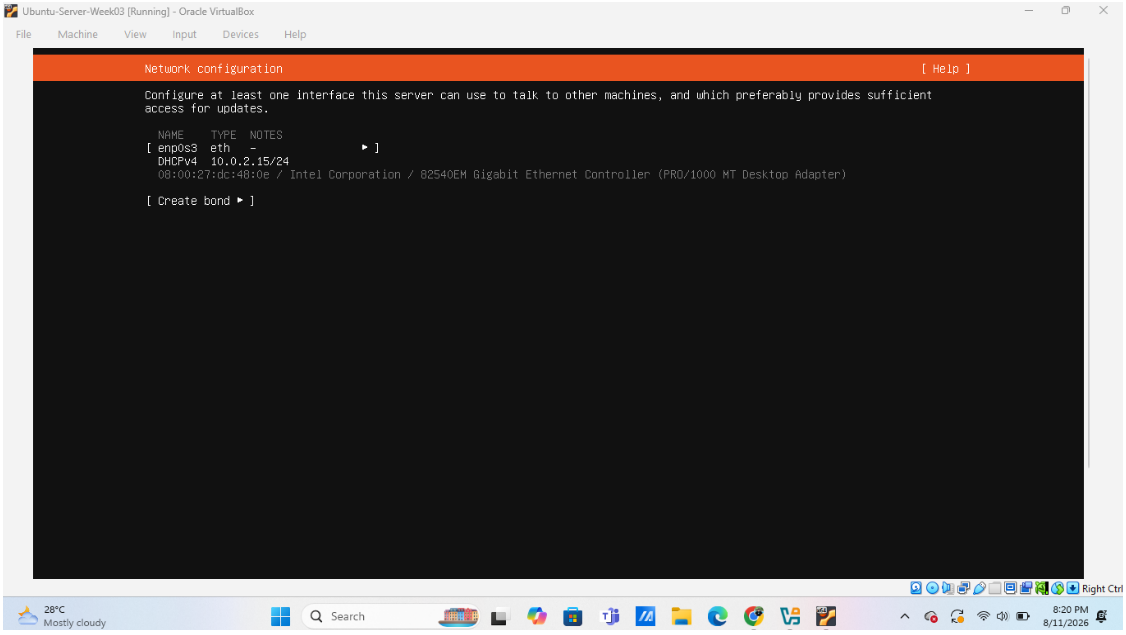
1. Choose Installation Type

At the base installation screen, “Ubuntu Server” (the default, full-featured option) was selected rather than the minimized variant, since the server will host multiple roles (file sharing, web hosting, database hosting).



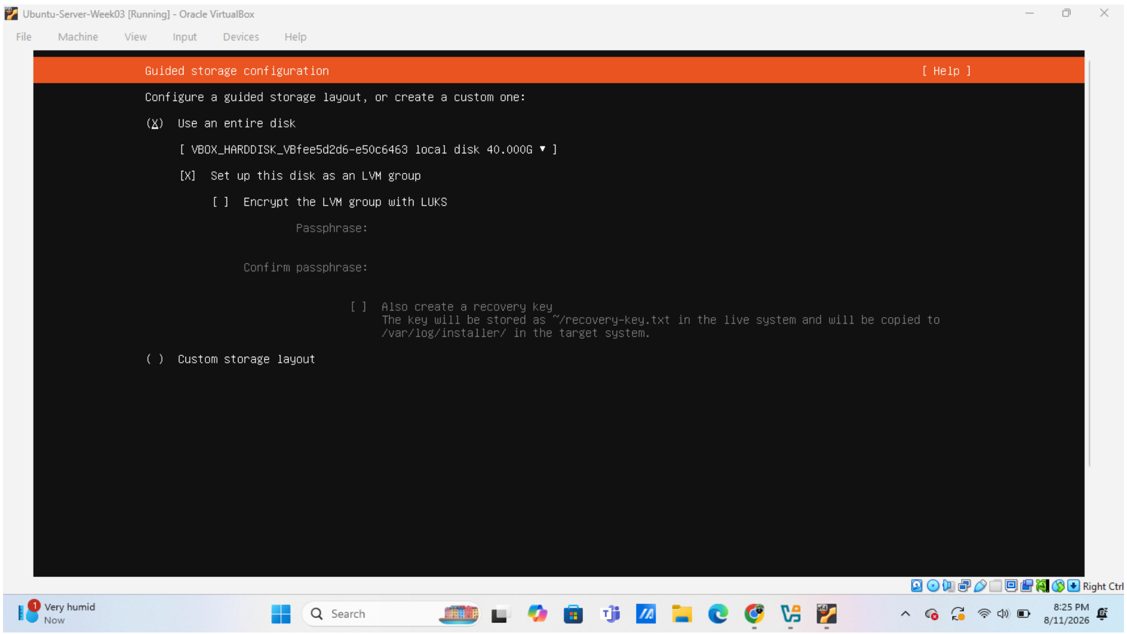
2. Network Configuration

The installer detected the network interface (enp0s3) and automatically assigned an IP address via DHCP, as instructed. The assigned address was recorded for later verification.

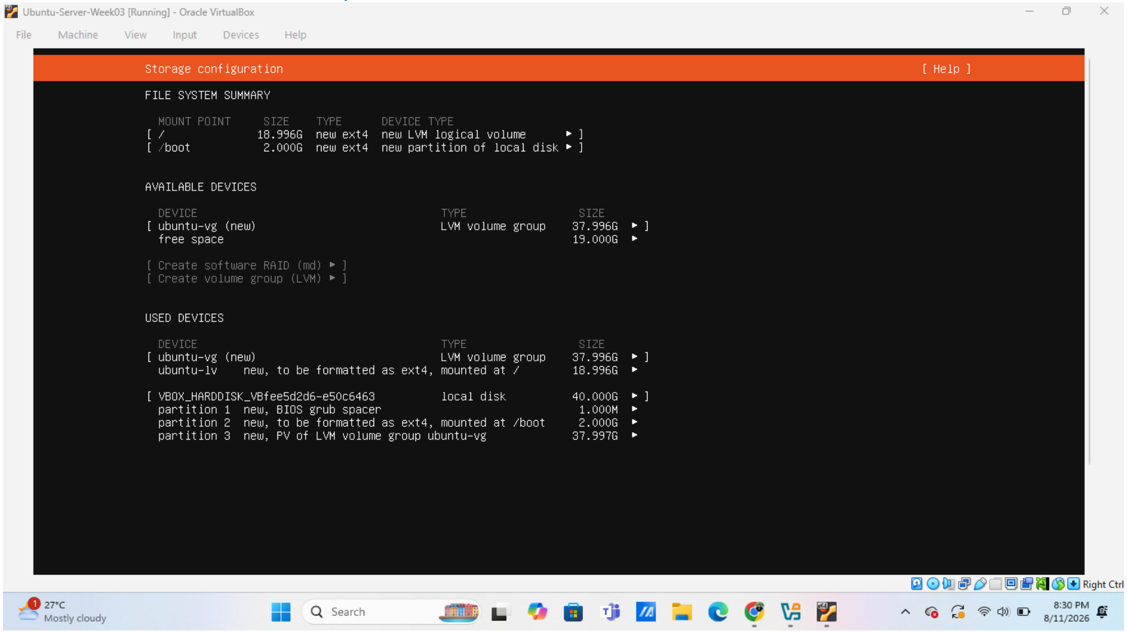


3. Disk Partitioning

Guided storage configuration was used with "Use an entire disk," configured as an LVM group for flexibility in future resizing. LUKS encryption was left disabled since it was not required for this lab environment.



The resulting partition scheme uses ext4 as the filesystem for both the root and boot partitions:

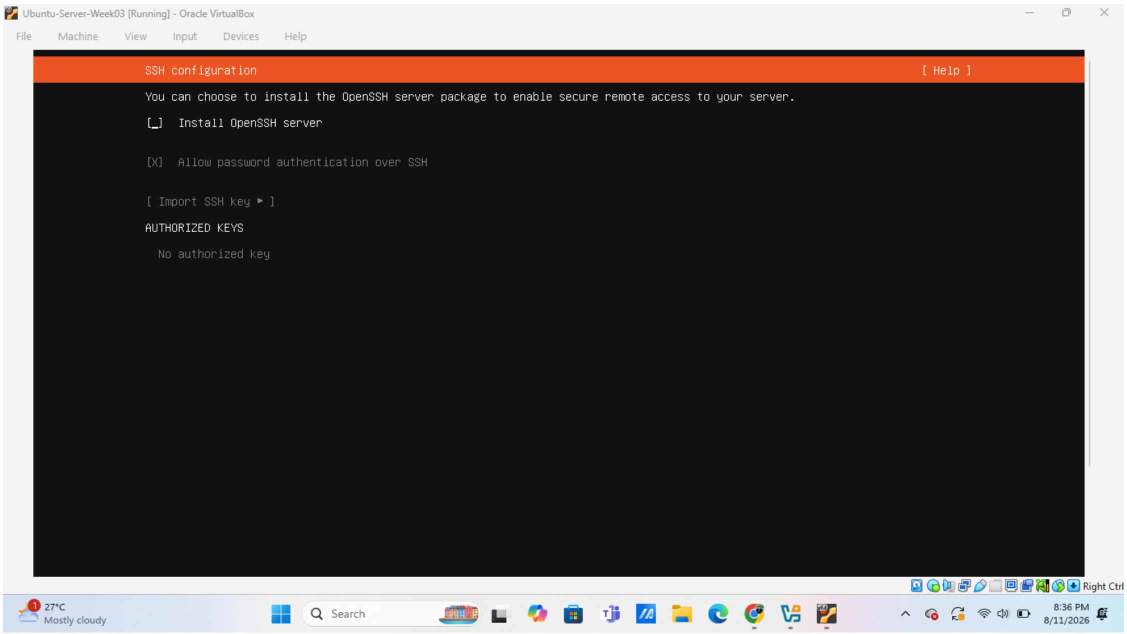


4. & 5. Hostname and User Account

The server was named “server01” and a non-root administrative user was created with a strong password, matching company standards for naming and access control.

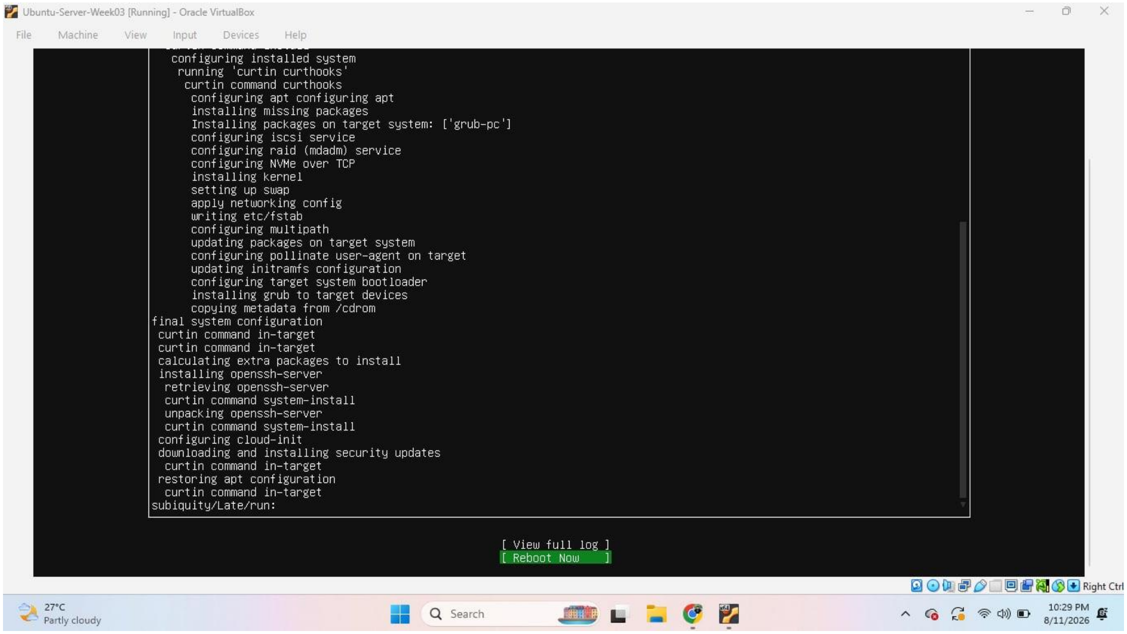
6. SSH Server

OpenSSH Server was enabled during installation to allow secure remote administration, a core requirement for this deployment.



7 & 8. Additional Packages and Installation

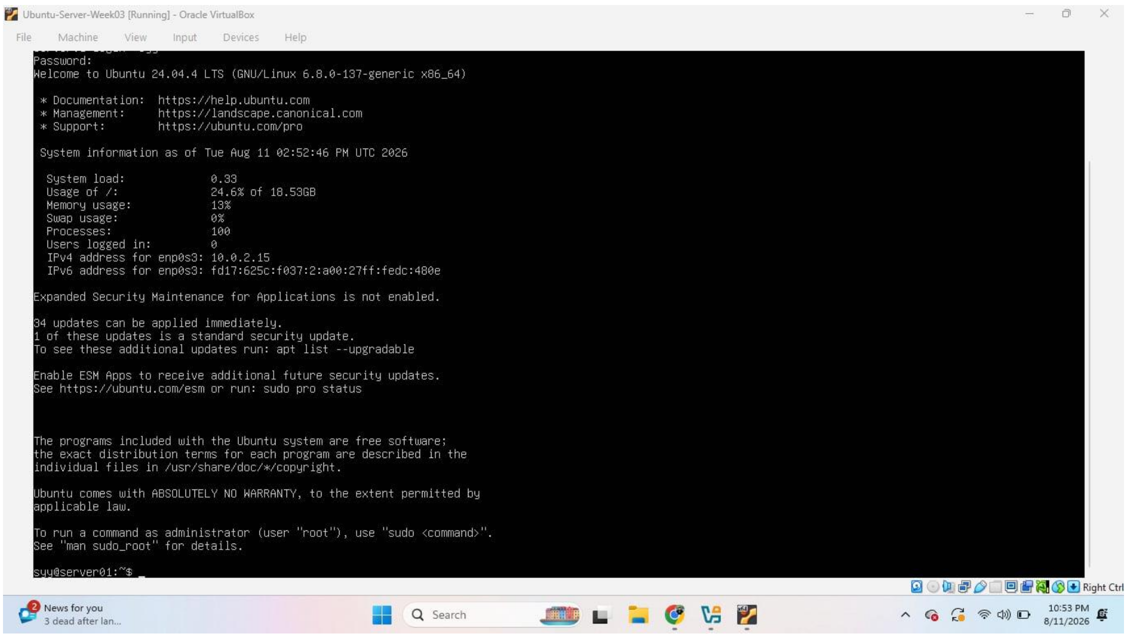
No additional packages/snaps were selected, per company instructions to avoid unnecessary software on a production server image. The installer then proceeded through package installation and configuration.



Server Verification

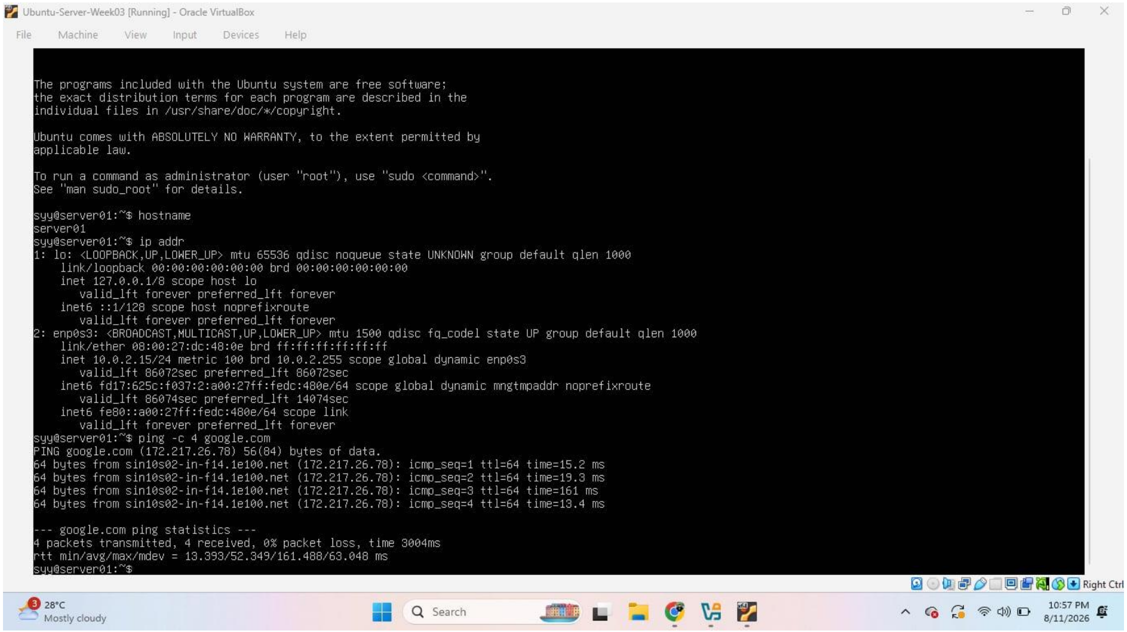
After installation and reboot, the following tasks confirmed the server was functioning correctly.

Task 1 – Login



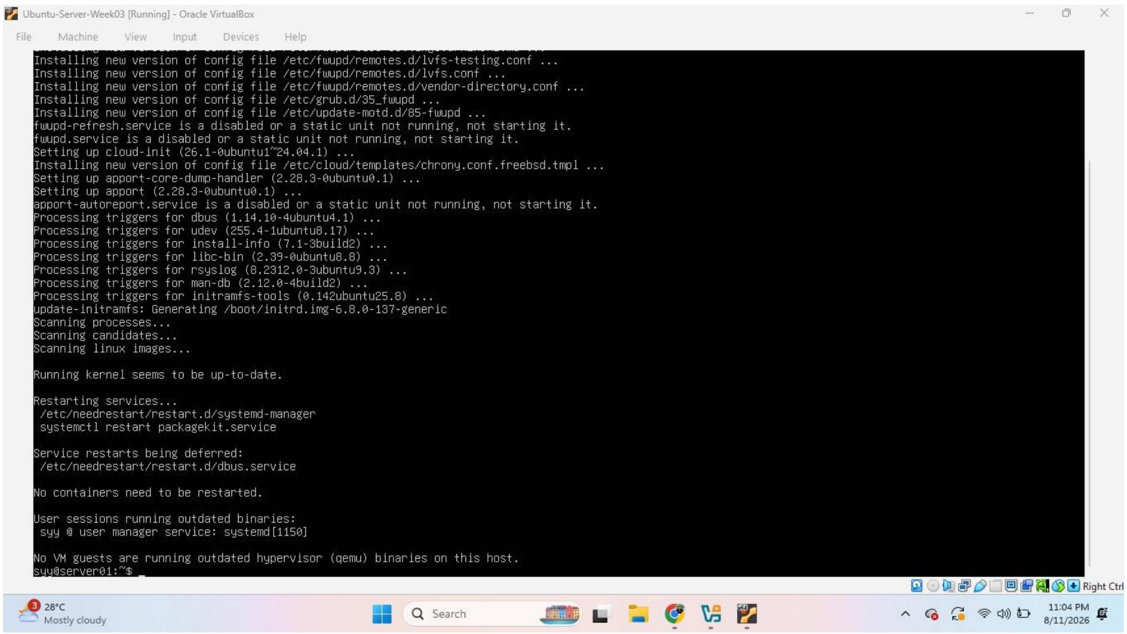
Tasks 2–4 – Hostname, IP Address, and Internet Connectivity

hostname ip addr ping -c 4 google.com



Task 5 – Update the Server

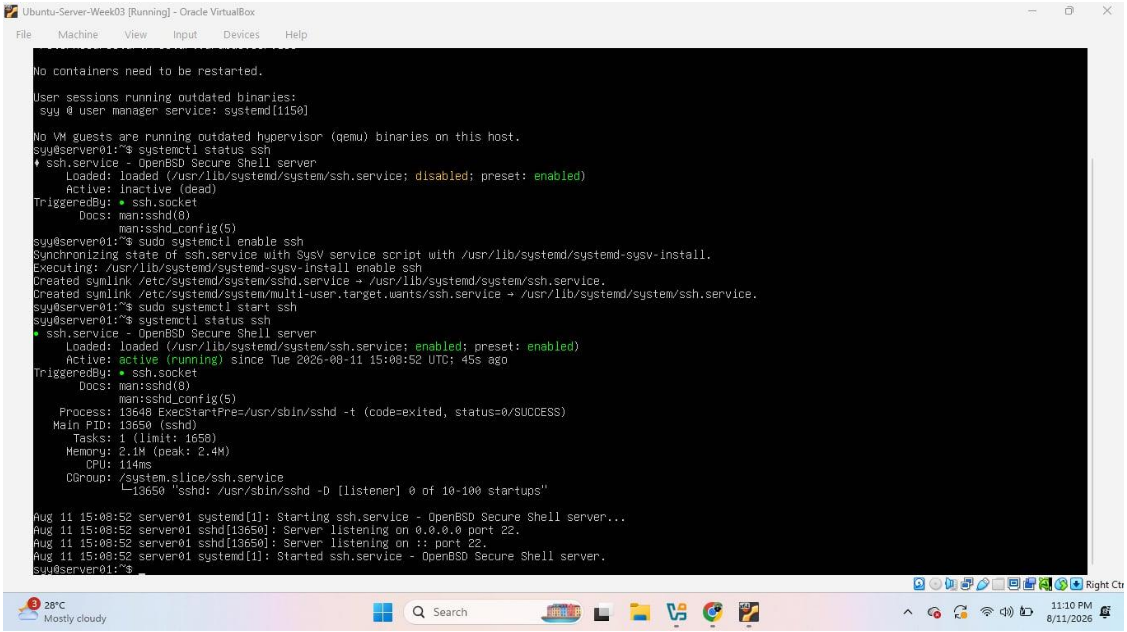
sudo apt update sudo apt upgrade -y



Task 6 – Verify SSH Service

The SSH service was initially found inactive despite being installed. It was enabled and started manually, then re-verified as active and running (see Problems Encountered for details).

sudo systemctl enable ssh sudo systemctl start ssh systemctl status ssh



BIOS vs UEFI Comparison

Category	BIOS	UEFI
Definition	Basic Input/Output System — firmware that initializes hardware and boots the OS in 16-bit real mode	Unified Extensible Firmware Interface — modern replacement for BIOS, runs in 32/64-bit mode
Boot Process	Reads the Master Boot Record (MBR) from the first disk sector	Uses the GUID Partition Table (GPT) and reads boot files from the EFI System Partition
Max disk Support	Up to 2 TB (MBR limitation)	Larger than 2 TB (theoretically up to 9.4 ZB)
Partition Style	MBR — max 4 primary partitions	GPT — supports up to 128 partitions
Security Features	Non built in	Secure Boot — blocks unauthorized/malicious bootloaders

Speed	Slower — sequential hardware initialization	Faster — supports parallel initialization, quicker POST
Advantages	Simple, broadly compatible with older hardware	Faster boot, more secure, supports large drives, GUI + mouse, network boot
Disadvantages	Slow boot, 2 TB limit, no security features, outdated	More complex, occasional compatibility issues with very old Oses
Modern Usage	Legacy systems, older PCs, some embedded devices	Standard on virtually all PCs, servers, laptops since ~2012+

Why UEFI Has Largely Replaced BIOS

The UEFI system has become the way that computers start up because it fixes some big problems with the old BIOS system. The BIOS system only works with a bit of memory and it is pretty slow. This is because it uses 16-bit mode, which's not very good for modern computers. The UEFI system is better because it uses 32-bit or 64-bit mode, which makes it faster.

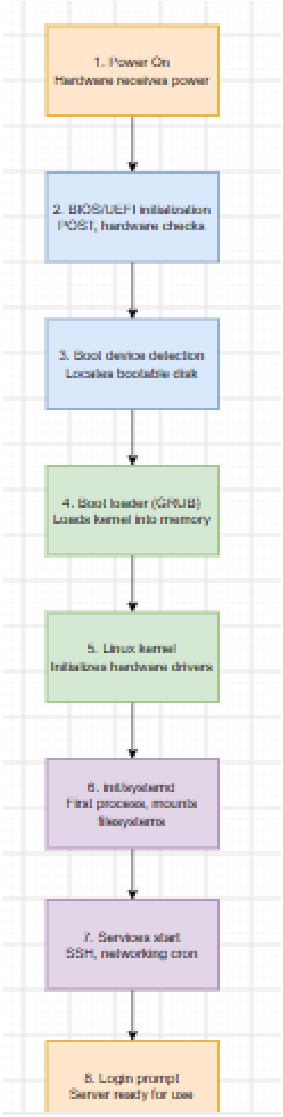
The old BIOS system also has some limitations. For example it uses something called MBR partitioning, which means that disks can only be a size. 2 TB. It also means that you can only have 4 partitions on a disk. The UEFI system uses something called GPT, which's much better. It lets you have disks that're much bigger and you can have up to 128 partitions.

The UEFI system is also more secure than the BIOS system. It has something called Secure Boot, which checks that all the software that loads when you start your computer is safe. This protects you from programs that might try to harm your computer. The BIOS system does not have this protection.

The UEFI system also has a modern interface, which is easier to use. You can use a mouse. Connect to the internet when you are setting up your computer. The UEFI system also starts up faster, than the BIOS system. Because of all these advantages all new computers and servers use the UEFI system. The BIOS system is now seen as technology.

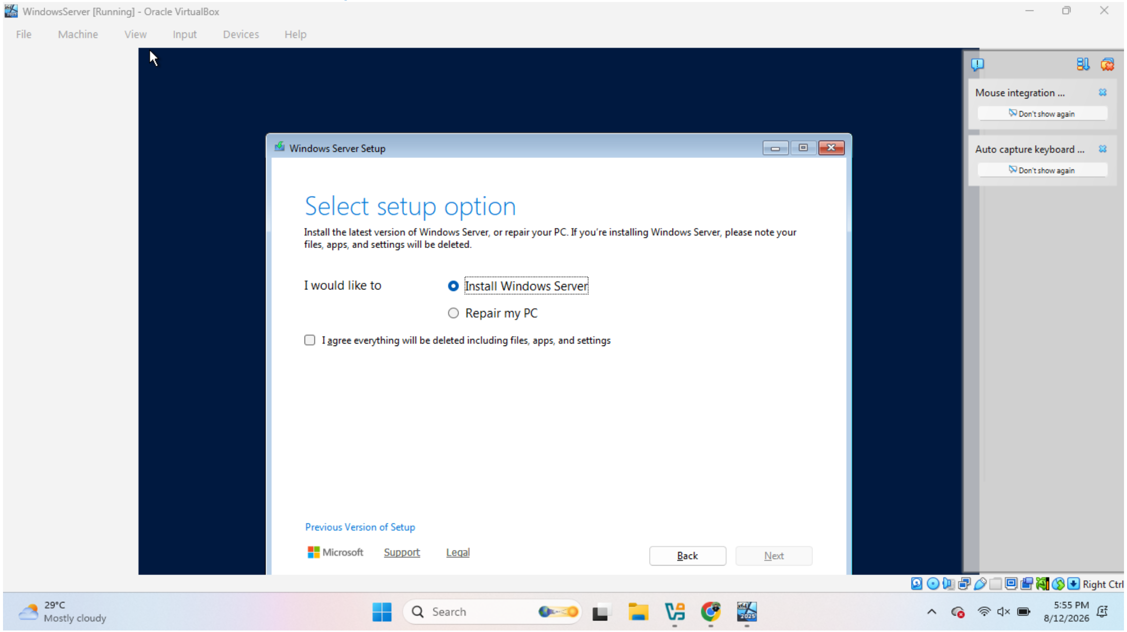
Boot Process Flowchart

The diagram below illustrates the Ubuntu Server boot sequence from power-on to login prompt: Power On → BIOS/UEFI Initialization → Boot Device Detection → Boot Loader (GRUB) → Linux Kernel → init/systemd → Services Start → Login Prompt.

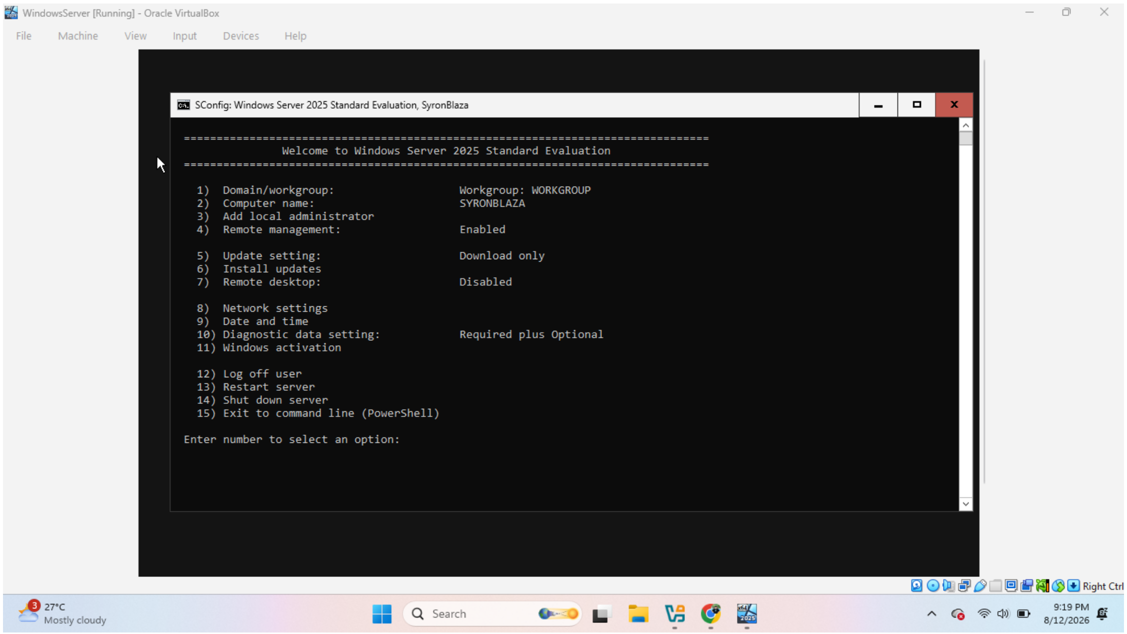


Windows Server Installation Summary

Windows Server 2025 Standard Evaluation was installed in a separate VirtualBox VM (32 GB disk, 1536 MB RAM) using the Server Core option (no desktop GUI) to accommodate limited host hardware resources. Setup followed the standard Windows Server Setup wizard: language/edition selection, license acceptance, disk selection, automated file copy and installation (with several automatic reboots), and first-logon password creation for the built-in Administrator account.



After installation, the server booted into SConfig, the text-based Server Configuration utility used to manage Server Core installations — confirming a successful install, computer name assignment, and administrator login.



OS Comparison Table – Windows Server vs. Ubuntu Server vs. Rocky Linux

Category	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Paid, per-core licensing (Standard/Datacenter); free 180-day evaluation	Free and open-source; optional paid Ubuntu Pro support	Free and open-source; no paid tier required, RHEL-compatible
User Interface	GUI (Desktop	CLI-only by default,	CLI-only by default,

	Experience) or Server Core (CLI-only)	managed via terminal/SSH	managed via terminal/SSH
Package Management	PowerShell, Windows Update, MSI/EXE, winget	APT (apt install / apt update) — Debian-based	DNF/YUM (dnf install) — Red Hat-based
Security	Windows Defender, Active Directory, Group Policy	AppArmor, UFW firewall, regular APT security patches	SELinux (strict mandatory access control), firewalld
Performance	Heavier resource usage, especially with GUI enabled	Lightweight, efficient on lower-spec hardware	Lightweight and stable, tuned for long-term enterprise use
Advantages	Deep Microsoft integration, strong enterprise support	Huge community, most popular cloud/server Linux distro, free	Very stable, 10-year support lifecycle, free CentOS alternative
Disadvantages	Costly licensing, higher resource requirements	Less “one-stop” enterprise support without a paid plan	Smaller community, less beginner documentation

Windows Server suits Microsoft-centric enterprises willing to pay for licensing and support. Ubuntu Server suits cloud-native, cost-conscious, DevOps-focused teams. Rocky Linux suits enterprises needing RHEL-grade stability and compliance without RHEL's licensing cost. (See the standalone Comparison Report for the full discussion.)

Problems Encountered

- VirtualBox refused to save VM settings with 4096 MB RAM allocated: “More than 80% of the host computer's memory is assigned to the virtual machine.” The host only had 3.5 GB of total RAM.
- The Ubuntu Server VM experienced a kernel soft lockup (“CPU#0 stuck for 58s”) during installation due to host CPU/RAM exhaustion from other running applications.
- After a VirtualBox restart, the VM disappeared from VirtualBox Manager with the error “Failed to open virtual machine … which has the same UUID as an existing virtual machine” — caused by a stale/duplicate entry in VirtualBox's registry file.
- The SSH service showed as installed but inactive (Active: inactive (dead)) immediately after first boot, rather than running as expected.
- The Windows Server installation progressed very slowly (multiple hours) due to the same constrained host hardware.

Solutions Implemented

- Reduced VM Base Memory from 4096 MB to 1536–2048 MB for both the Ubuntu and Windows Server VMs, bringing allocation under the host's safe threshold.
- Closed unnecessary host applications and browser tabs to free CPU and RAM before

and during installation, allowing the stuck installation to resume and complete.

- Resolved the duplicate VM registration by locating C:\Users\<user>\.VirtualBox\VirtualBox.xml, backing it up, removing the stale <MachineEntry> line with the mismatched UUID, and re-adding the VM via File → Add Machine.
- Enabled and started the SSH service manually with sudo systemctl enable ssh and sudo systemctl start ssh, then confirmed it was active (running).
- Allowed the Windows Server installation to run uninterrupted in the background rather than cancelling it, since progress — though slow — was continuing.

Lessons Learned

- Always check host hardware specifications (RAM/CPU) before allocating VM resources; assignment “minimum requirements” may need to be adjusted for the actual lab hardware available, and that adjustment is worth documenting rather than hiding.
- Services marked as “installed” during setup are not always “active” after first boot — always verify with systemctl status rather than assuming installation implies a running state.
- VirtualBox's VM registry (VirtualBox.xml) can desync from the actual VM files on disk after a crash; the underlying .vdi/.vbox files usually survive even when the Manager UI loses track of them, so a crash is rarely true data loss.
- Low-resource host environments make Server Core / CLI-only installations (no desktop GUI) the more practical choice for both Linux and Windows Server, since GUI environments consume RAM that CLI administration does not need.
- Patience and incremental verification (checking one task at a time with screenshots) made it much easier to diagnose exactly where and why something failed, compared to running many steps before checking results.

References

Ubuntu Server Documentation – https://ubuntu.com/server/docs
Oracle VirtualBox User Manual – https://www.virtualbox.org/wiki/Documentation
Microsoft Windows Server 2025 Evaluation Center – https://www.microsoft.com/evalcenter/download-windows-server-2025
Rocky Linux Documentation – https://docs.rockylinux.org